Raj Yadav

Python Backend & Applied AI Engineer

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Summary

Python backend and applied AI engineer with a B.Tech in Computer Engineering and an MBA (Finance) from NMIMS Mumbai (2025). I build REST APIs with FastAPI, generative AI applications with LLMs, RAG and LLM evaluation, data pipelines, and event-driven systems on Docker, Kubernetes, AWS and GCP. I build with AI coding agents (Claude Code, Codex) and own the design, review and verification of everything I ship.

Skills

- **AI & LLMs:** LLMs, Generative AI, RAG, LLM evaluation, prompt engineering, embeddings, vector search, open-source LLMs (Ollama), Anthropic Claude API, AI coding agents (Claude Code, Codex)
- **Machine learning:** scikit-learn, XGBoost, LightGBM, statsmodels, SciPy, deep learning (TCN and causal 1D CNN models implemented in C++), MLOps (model registry, artifact verification, shadow validation), statistics (FDR / Holm testing, bootstrap, walk-forward validation)
- **Backend & data:** Python, FastAPI, REST APIs, WebSockets, Pydantic, SQLAlchemy, SQL, PostgreSQL, Redis Streams, DuckDB, pandas, NumPy, ETL / data pipelines, data modeling, system design, event-driven and distributed systems (outbox pattern, idempotent consumers, dead-letter queues)
- **Cloud & DevOps:** Docker, Kubernetes, AWS, GCP (Google Cloud), CI/CD (GitHub Actions), Linux, Git, pytest, Prometheus, Grafana
- **Also:** C++20, CMake, TypeScript / JavaScript, React, Playwright, Power BI, Advanced Excel, SAP

Projects

Algo Matrics: multi-tenant algorithmic trading SaaS | algomatrics.in

Python, FastAPI, PostgreSQL, Redis, Docker, Kubernetes, AWS, GCP, React, TypeScript, Claude API

- Built a multi-tenant SaaS with REST APIs in FastAPI: plans and billing (Razorpay / Stripe with GST), broker connections (Zerodha, Angel One, Flattrade), risk limits and kill switches.
- Runs across AWS and Google Cloud: a GCP trading VM sends monitoring batches over HTTPS to the AWS platform; deploys with one Docker Compose command or on Kubernetes with KEDA / HPA autoscaling.
- Designed the system as a Clean Architecture / DDD monorepo with an event bus on Redis Streams (outbox relay, idempotent consumers, dead-letter handling); 152 commits, 219 test files, CI/CD in GitHub Actions, and 330 ops-backend, 440 platform and 118 frontend tests passing (Sep 2026).
- AI layer: an assistant on the Claude API that answers only from figures in its context, and an AI-CIO forecast whose top-10 hit rate reached 27.8% against an 8.2% base rate (3.4x lift) in backtest.

Research Copilot: RAG with a local open-source LLM

Python, Ollama, Qwen2.5, nomic-embed-text, BM25, NumPy, FastAPI

- Built retrieval-augmented generation over 299 project documents (5,126 chunks): hybrid retrieval combining BM25 and dense vector search over embeddings with reciprocal rank fusion, answered by Qwen2.5 running locally.
- Wrote an LLM evaluation harness on labelled questions: retrieval hit@5 100%, MRR 0.71, 64% of expected facts present, 2 of 2 off-topic questions refused.
- Added a grounding check that cites each answer sentence to its source passage and flags unsupported ones (92% of sentences grounded on average); secrets are redacted before indexing.

Intraday Research Lab: ML research on NSE options

Python, pandas, scikit-learn, XGBoost, SciPy, Linux, GCP

- Built data pipelines that rebuild 30-second and 1-minute candles from raw ticks, and engineered 1,705 features tested with walk-forward validation, FDR / Holm multiple-testing control and explicit trading costs.
- Ran a paper-trading runtime on a Google Cloud Linux VM; when the best strategy's edge (+₹15 per trade) turned negative at the measured 733 ms execution delay, the research recommended committing no

capital.
- 330,000+ lines of Python and C++ across 1,458 files, with 355 test files.

LLS: low-latency trading infrastructure in C++20

C++20, CMake, Python

- Lock-free SPSC ring-buffer pipeline from the market feed to a preallocated market-state engine with zero-copy snapshots.
- Deep learning and MLOps in C++: TCN and causal 1D CNN models, tree ensembles, purged walk-forward splits, leakage detection, a model registry with verified artifacts, shadow validation and a zero-allocation inference engine.
- A differential oracle proves the C++ port reproduces the Python strategy exactly; 214 source and 98 test files with ASan / UBSan builds.

RQIE: quantitative research engine

Python, DuckDB, pandas, statsmodels, scikit-learn

- Pre-registered experiments across 27 sessions of NSE option data, on a DuckDB data store with per-session verification and an independent store audit.
- An execution-cost model and delay stress tests showed no edge survived real trading costs (statutory charges about 0.24% per trade), so none was claimed.

StudyPilot: offline-first study app | studypilot-biw.pages.dev

React, TypeScript, PostgreSQL (Supabase), PWA

- Live app covering 9 textbooks, 256 topics and 1,232 practice questions, with timed papers that save every answer and submit exactly once.
- Offline-first: an IndexedDB outbox with idempotency keys syncs finished work when the connection returns.

Experience

Management Intern, Tata Capital Housing Finance (TCHFL) | May 2024 – Jul 2024

- Automated MIS reports and built dashboards for senior management.
- Audited P&L statements from GL files in Excel and analyzed key financial performance metrics and trends.

Technical Intern, NTPC Ltd. | May 2023 – Jul 2023

- Analyzed energy trade patterns in Power BI and SAP to evaluate a potential demerger of the renewable and coal businesses.
- Contributed to the team's final report, after which the demerger idea was dropped.

Education

MPSTME, NMIMS Mumbai | 2025
MBA Tech: B.Tech Computer Engineering + MBA (Finance major, Business Intelligence & Analytics minor) | CGPA 2.65 / 4

Certifications

- Trading Algorithms, Indian School of Business (Coursera)
- Advanced Trading Algorithms, Indian School of Business (Coursera)
- Investment and Portfolio Management, Rice University
- Crash Course on Python, Google
- Learn Python 3, Codecademy

Achievements

- Winner, Smart India Hackathon 2022
- Finalist, Startup India: India's Social Impact Champion 2023
- Third place, audio/video summarization, Saavishankar competition 2024